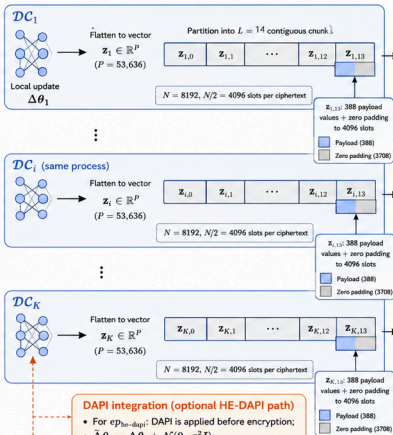
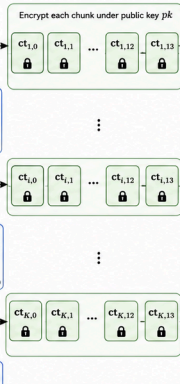


CKKS Secure Aggregation Pipeline

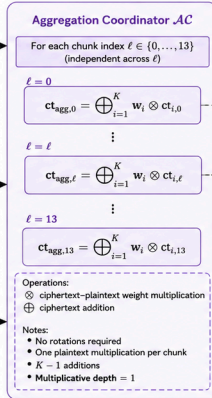
1 Client-side gradient packing



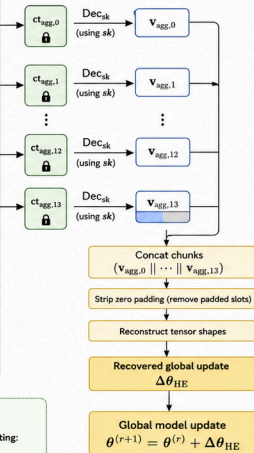
2 CKKS encryption



3 Ciphertext-domain weighted aggregation



4 Decryption, unpacking, and global update



DAPI integration (optional HE-DAPI path)

- For $ep_{he-dapi}$: DAPI is applied before encryption;
 - $\tilde{\Delta}\theta_i = \Delta\theta_i + \mathcal{N}(0, \sigma_i^2 I)$;
 - Weights w_i are derived from trust score T_i .
- $\tilde{\Delta}\theta_i$, not raw $\Delta\theta_i$, is packed and encrypted in the HE-DAPI path.

CKKS context parameters

- CKKS context: $N = 8192$
- coeff_mod = $[60, 30, 60]$
- $\Delta = 2^{30}$
- depth = 1

Parameter selection (why this setting)

Constraints:

- C1: 128-bit RLWE security
- C2: depth-1 aggregation
- C3: $\delta_{agg} \leq K \cdot 2^{-30} < 10^{-6}$

Evaluated setting:

- $K = 5$ clients
- $\delta_{agg} < 10^{-6}$

Legend

→ Data flow

--- Optional DAPI path

Symbols: pk public key, sk secret key